COMPUTER VISION WITH DEEP LEARNING · FINAL PROJECT

# TriFrame CV

## Can pictures of stock charts predict risk?

500 US stocks · 236,520 chart sets · 7 pretrained vision networks · 2001–2022




Krishna Harsha Tangirala · github.com/harshatangirala/triframe-cv · October 2026

THE ANSWER IN ONE SLIDE

# Risk is predictable. Direction is not.

### 20-DAY VOLATILITY

**0.503**

$R^2$ on the 2020–2022 test years

Explains half the variation in future volatility and beats HAR, EWMA and rolling volatility.

### 10% DRAWDOWN

**0.648**

ROC-AUC on the test years

Ranks a stock that later falls 10% above one that does not 65% of the time.

### 5-DAY DIRECTION

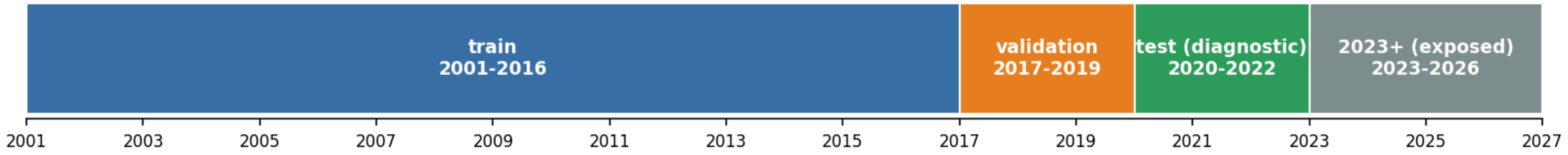
**0.510**

ROC-AUC on the test years

A coin flip. The decision layer therefore defaults to NO TRADE.

Chart images add no significant gain over the 33 numeric features on the test years: $\Delta R^2$ +0.003 (95% CI −0.018 to +0.026).

# 236,520 chart sets from 500 US stocks




| Period | Years | Chart sets | Stocks | 10% drawdowns |
|---|---|---|---|---|
| Train | 2001–2016 | 165,421 | 489 | 22.7% |
| Validation | 2017–2019 | 35,145 | 492 | 17.8% |
| Test | 2020–2022 | 35,954 | 500 | 30.2% |
| 2023 onward | exposed | – | – | unused |

**979**
US stocks with daily and hourly OHLCV bars

**500**
sampled at random; they contain the 60- and 120-stock runs

**1,913**
rows purged where a 20-day label crossed a split

# Three text-free charts per stock-day




### Renderer

OHLC bars, 3 px each, 128 px high: a 100 px price panel over a 26 px volume panel. No text, no gridlines.

### Network input

Pixels copied to 3 channels, resized to 224 × 224 with antialiasing and normalised with each network's ImageNet statistics.

### The key consequence

Log prices are stretched to fill each panel, so a picture shows the shape of moves, never their size.

# Three targets, 33 numbers, one window

**V** **20-day volatility**

Log realised volatility of daily returns over the next 20 sessions

`regression · $R^2$, RMSE`

**D10** **10% drawdown**

Price trades 10% or more below the next day's open within 20 sessions

`yes/no · base rate 22.7 / 17.8 / 30.2%`

**05** **Beats the median stock**

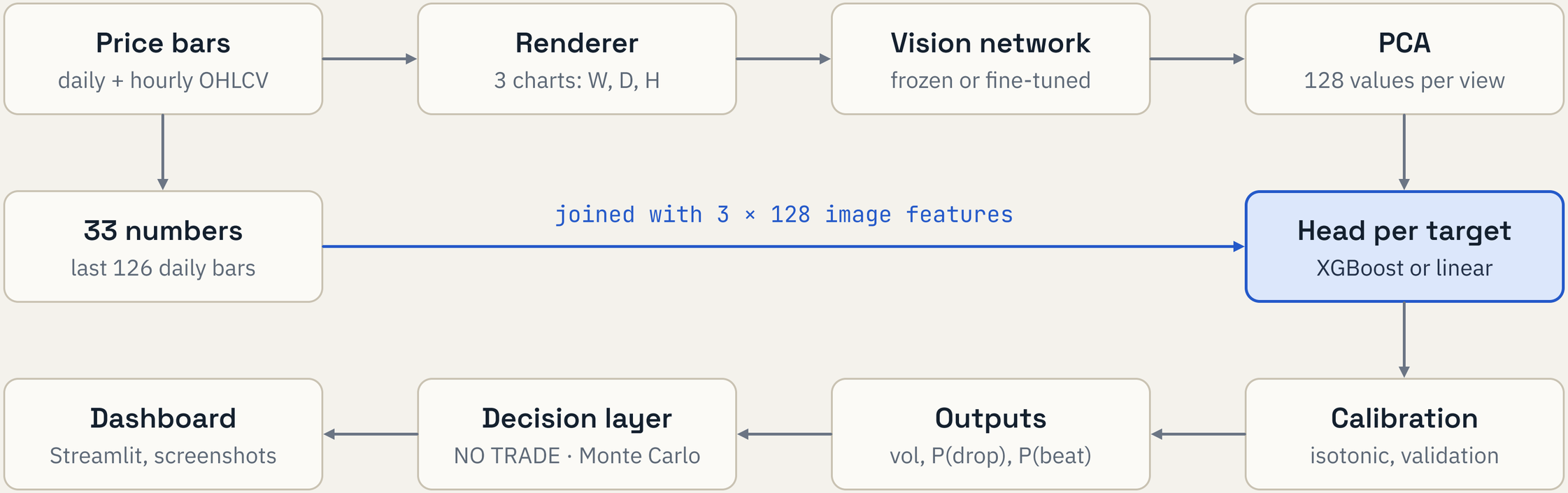
5-day return above the same-date median of the universe

`yes/no · base rate 50%`

| Group | The 33 numeric features |
|---|---|
| Returns | 1-, 5- and 21-day returns; overnight vs intraday split; gap |
| Volume | abnormal volume; 5-day volume surge |
| Mean reversion | RSI (2, 3, 14); 20-day z-score; streaks; variance ratio; AR(1) |
| Volatility | close-to-close, Parkinson and Garman-Klass at 1, 5 and 22 days; EWMA |
| Amplitude | ATR %, log range, largest drawdown in the window |

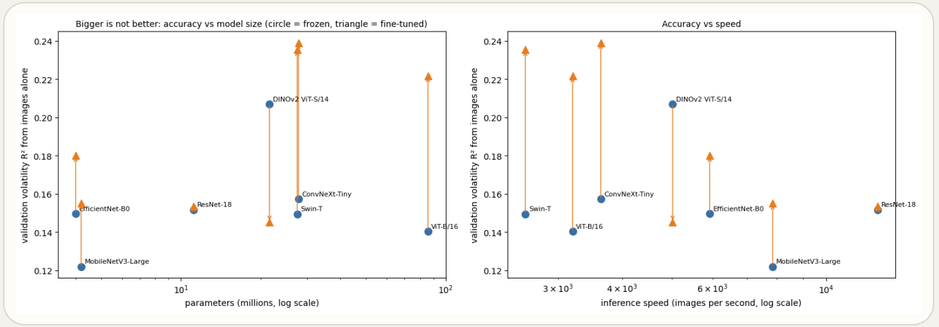
All 33 come from the last 126 daily bars, the same window a daily screenshot shows. Gaps are filled with training medians; scaling is fitted on training years only.

# Images and numbers meet in one head



# Seven networks, three random controls

| Network | Type | Pretraining data | Params |
|---|---|---|---|
| ViT-B/16 | Transformer | ImageNet-21k → 1k | 85.8 M |
| ConvNeXt-Tiny | CNN | ImageNet-12k → 1k | 27.8 M |
| Swin-T | Transformer | ImageNet-1k | 27.5 M |
| DINOv2 ViT-S/14 | Transformer | LVD-142M, self-sup. | 21.6 M |
| ResNet-18 | CNN | ImageNet-1k | 11.2 M |
| MobileNetV3-L | CNN | ImageNet-1k | 4.2 M |
| EfficientNet-B0 | CNN | ImageNet-1k | 4.0 M |
| ResNet-18 × 3 | Control | none (random) | 11.2 M |




Bigger is not better: $R^2$ from images alone against size and speed.

**DINOv2 (21.6 M)** reads the most from pictures alone: $R^2$ 0.207. Random-weight ResNets: about −0.02.

Heads: ridge, logistic and XGBoost (depth 3–7, ≤ 2,000 trees). 1,881 configurations logged.

# Six safeguards against fooling ourselves

### Time-ordered splits

Train 2001–16, validate 2017–19, test 2020–22. 1,913 rows whose 20-day label crossed a boundary were purged.

### Train-only fitting

Gap filling, scaling and PCA learn from training rows only, so nothing from later years leaks backwards.

### Honest uncertainty

Overlapping 20-day labels are not independent, so every CI resamples whole calendar months (1,000 draws).

### Pre-registered rule

An image model wins only if its gain over numbers has a CI above zero and holds in every validation year.

### Negative controls

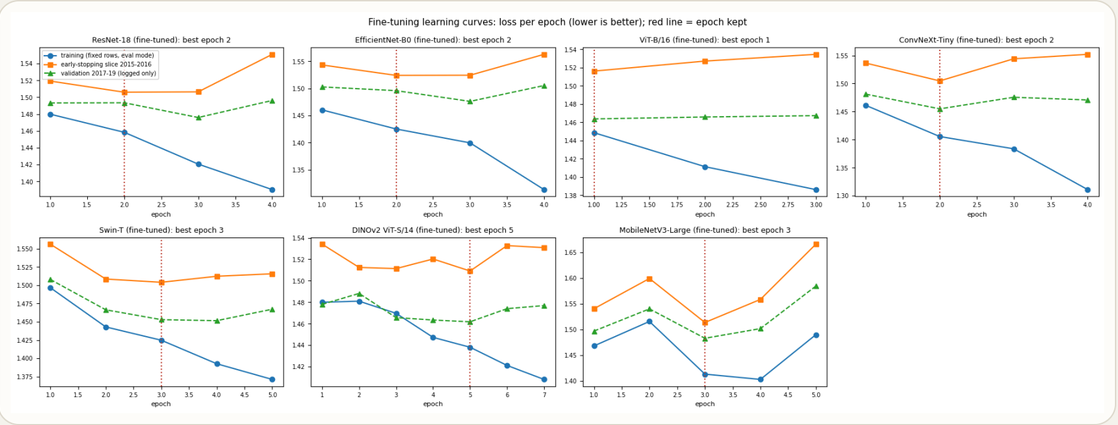
Shuffled labels give $R^2$ −0.123; random-weight networks about −0.02. Nothing is found where nothing exists.

### Provenance

Every metric carries code and config hashes; the deployed bundle matches the notebook to within $4 \times 10^{-5}$.

# End-to-end fine-tuning, validation untouched

| Setting | Value |
|---|---|
| Loss | joint MSE + BCE, 3 targets |
| Optimiser | AdamW, weight decay 0.05 |
| Learning rate | 3e-5 backbone · 1e-3 head |
| Schedule | 5% warm-up, cosine decay |
| Batch · precision | 64 · bfloat16 · A100 |
| Gradient clipping | norm 1.0 |
| Data per epoch | fresh 60,000 of 141,525 sets |
| Early stopping | 2015–16 slice, patience 2 |
| Max epochs | 8 |
| Augmentation | none |




Loss per epoch; the red line marks the epoch kept.

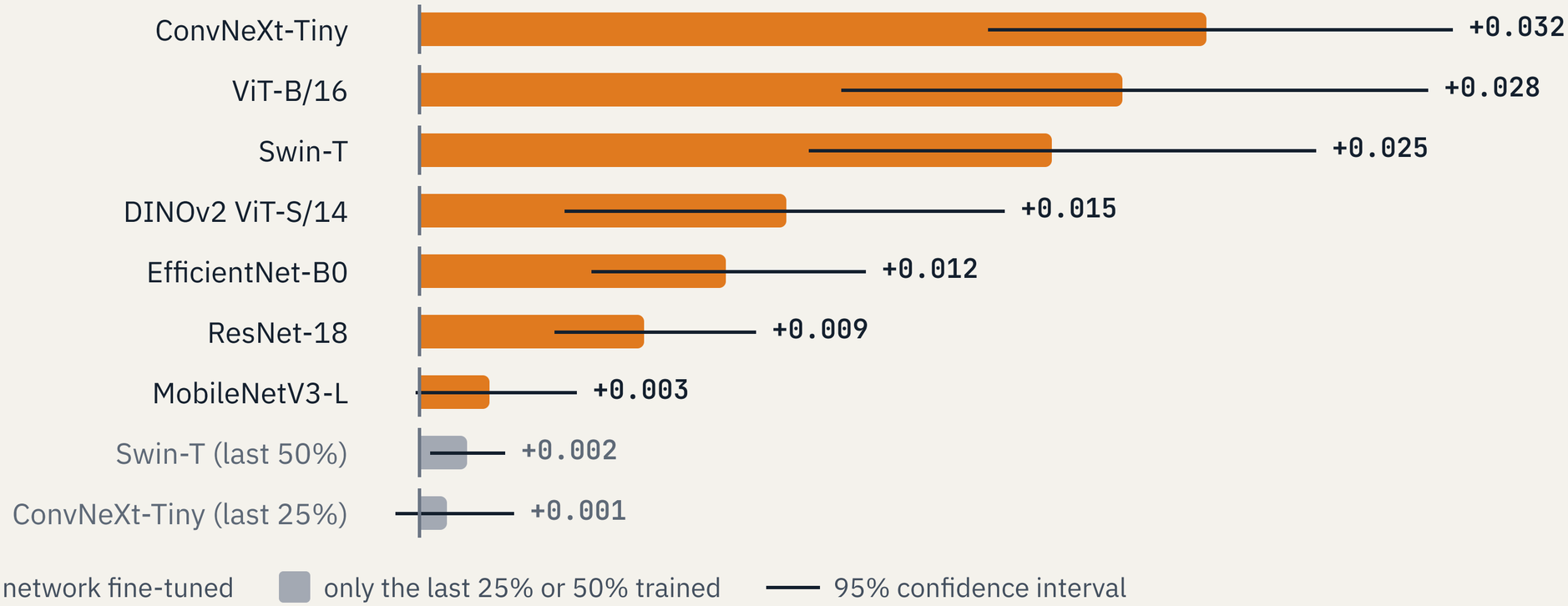
**15**
fine-tuning runs

**158**
GPU-minutes

**0**
validation rows used

# Fine-tuning works only on the whole network

Validation $R^2$ gain from fine-tuning (numbers + images) with 95% month-block CIs. 9 of 9 runs improved, 7 significantly.



# Convex heads: a certified global minimum

CERTIFIED GLOBAL MINIMUM

**Ridge head (volatility)**

Gradient at the solution $6.1 \times 10^{-16}$; 5 random starts reach it.

CERTIFIED GLOBAL MINIMUM

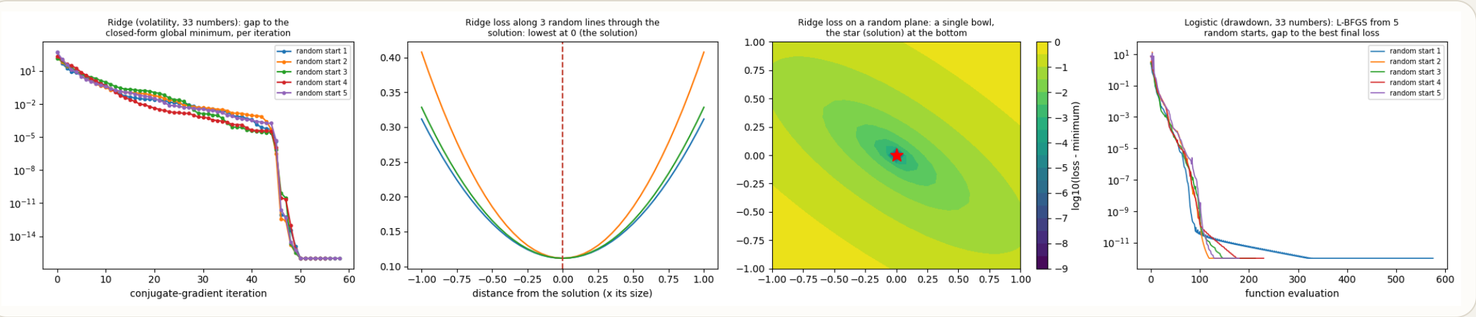
**Logistic head (drawdown)**

5 random starts agree in loss to $2.5 \times 10^{-14}$.

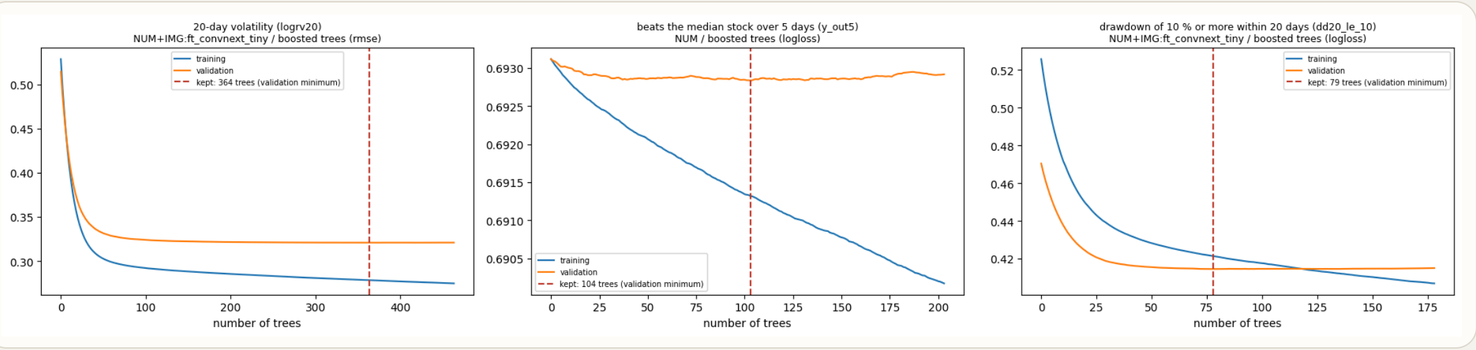
VALIDATION MINIMUM KEPT

**Boosted trees**

Greedy, not convex: kept at 364 (V), 79 (D10) and 104 (O5) trees.

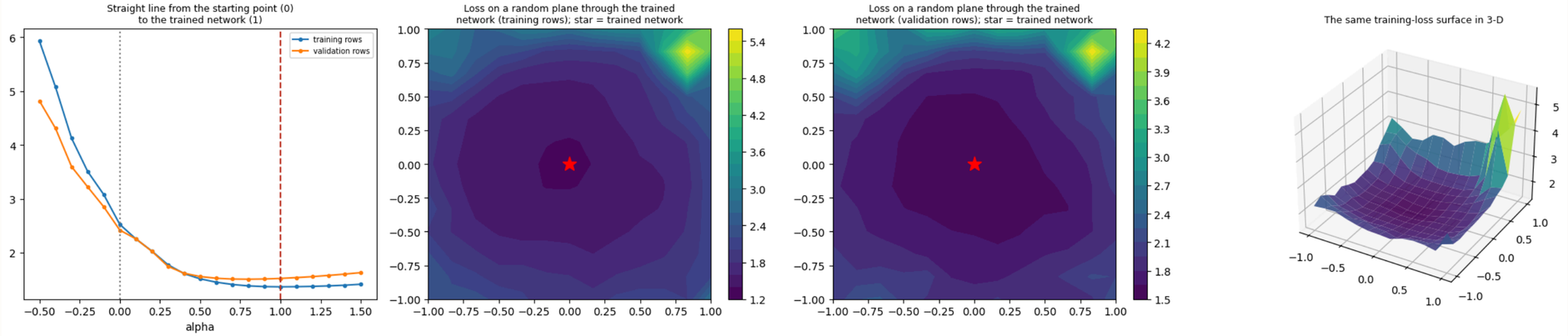



Five random starts, one minimum; the ridge loss is a single bowl.




Boosting curves: training keeps falling; the dashed line is the kept validation minimum.

# The network sits at the bottom of its valley




**1.461 → 1.311**
training loss over 4 epochs

**Epoch 2 of 4**
lowest held-out loss; that checkpoint is kept

**α = 1.0**
lowest point on the start-to-trained path

**0.007**
seed-to-seed SD of validation $R^2$

No method can certify a global minimum for a non-convex network. The target is the lowest held-out loss, not the lowest training loss, which would only memorise the training years.

# Scores on the 2020–2022 test years

| Target | Metric | Validation | Test (95% CI) | Numbers only, test |
|---|---|---|---|---|
| Volatility | $R^2$ | 0.592 | 0.503 (0.25–0.65) | 0.500 |
| Volatility | RMSE, log vol | 0.321 | 0.381 | 0.382 |
| Volatility | Spearman ρ | 0.758 | 0.742 | 0.726 |
| Drawdown | ROC-AUC | 0.740 | 0.648 (0.59–0.69) | 0.653 |
| Drawdown | Prec. / recall / F1 | 0.36 / 0.55 / 0.43 | 0.39 / 0.65 / 0.49 | 0.37 / 0.74 / 0.49 |
| Direction | ROC-AUC | 0.516 | 0.510 (0.500–0.521) | 0.510 |

### THRESHOLDS

Fixed on validation: 0.223 for drawdown, 0.439 for direction.

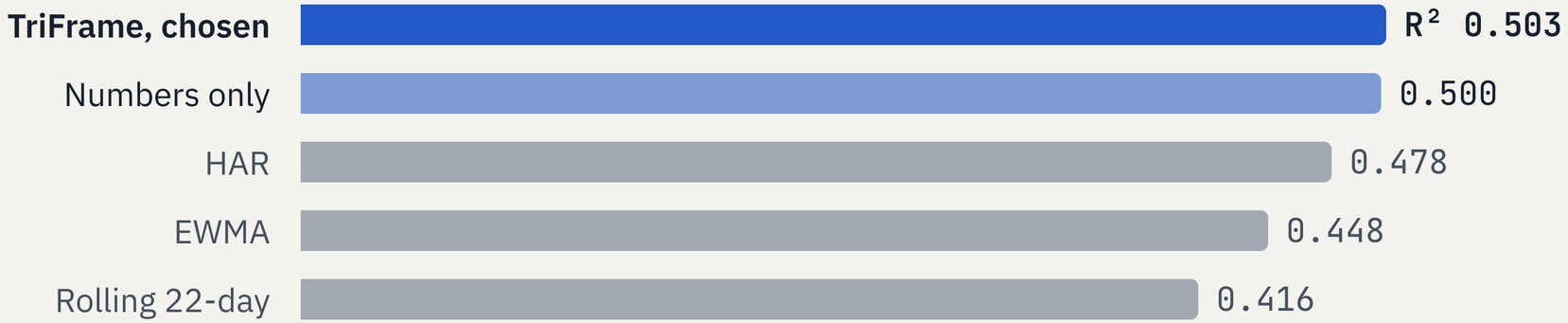
### DRAWDOWN, TEST

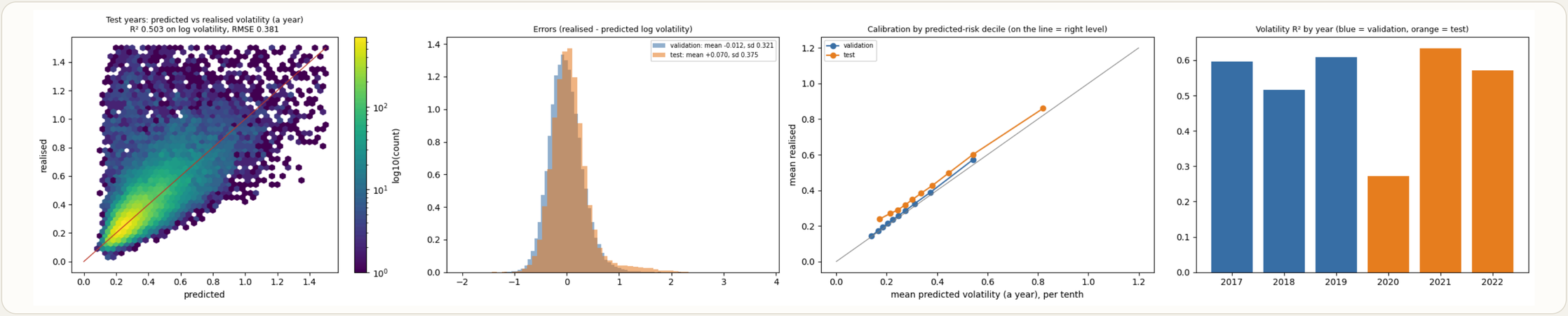
Balanced accuracy 0.607; Brier skill 0.046 over the base rate.

### STATUS OF THE TEST YEARS

Read by earlier runs, so a historical check, not a fresh test.

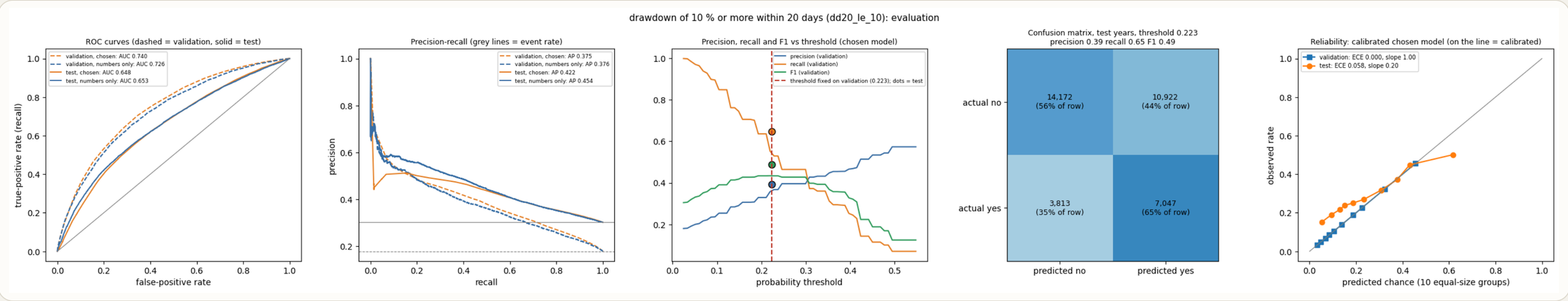
# Ahead of every classical volatility model




Test R² from zero. Below: predicted vs realised, errors, calibration by decile and R² by year (blue = validation, orange = test).

# Drawdown risk ranked well above chance




ROC and precision-recall curves, threshold curves, test confusion matrix and reliability (dashed = validation, solid = test).

**0.648**
ROC-AUC, 95% CI 0.59 to 0.69

**0.39**
precision against a 0.30 base rate: 1.3× lift

**0.65**
recall at the validation threshold of 0.223

**0.058**
calibration error: base rate rose from 17.8% to 30.2%

# The risk flag separates risky stocks

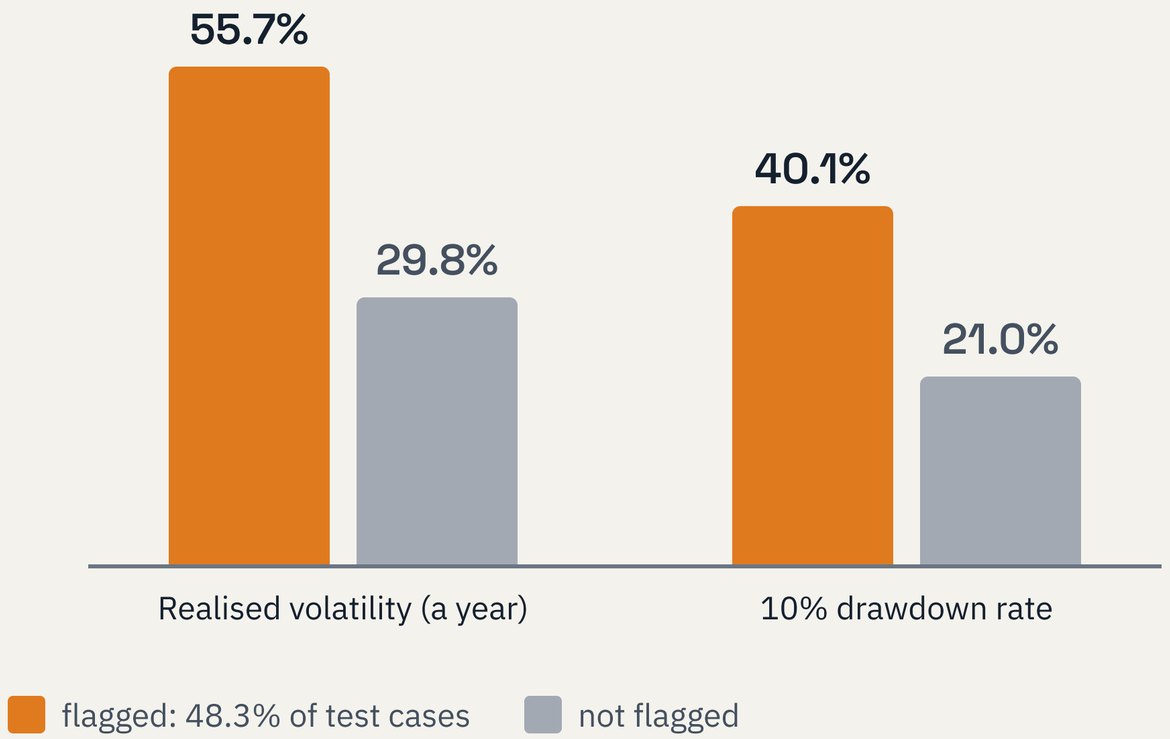
RULE FIXED ON VALIDATION

## Flag a stock as high risk when

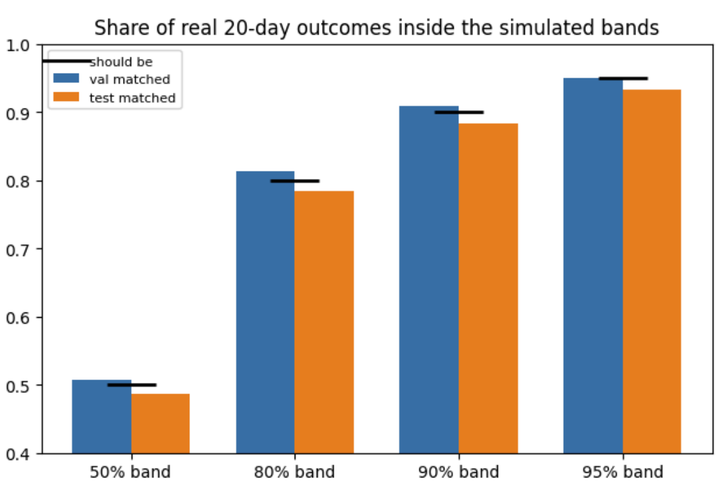
- predicted volatility ≥ 33.6% a year
- or P(10% drawdown) ≥ 30.3%

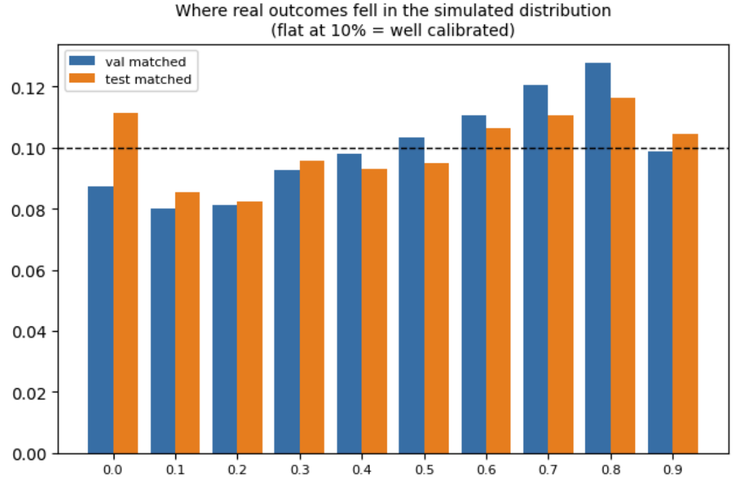
Both cut-offs are the top fifth of validation readings.

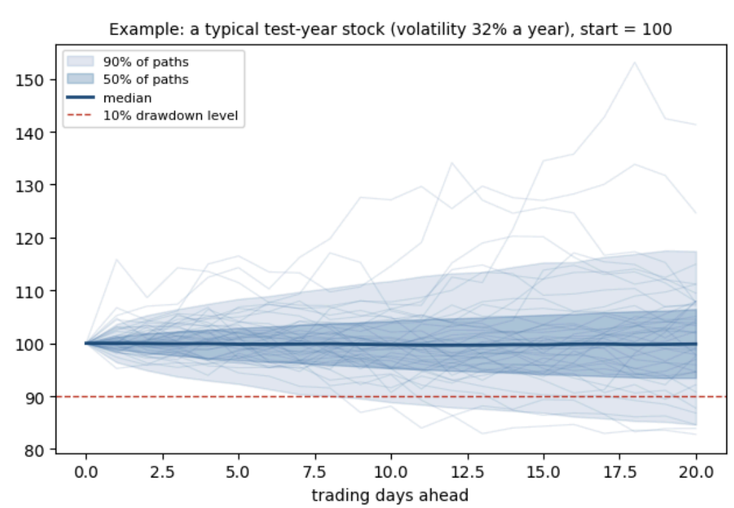
A direction call appears only if direction passes both validation and test checks. It did not, so every decision is **NO TRADE**.



# Simulated price paths, calibrated on validation




Band coverage against the target, where real outcomes fell in the simulated distribution, and an example fan of 20-day paths.

**88%**
of test outcomes inside the 90% band (validation 91%)

**49%**
inside the 50% band (validation 51%)

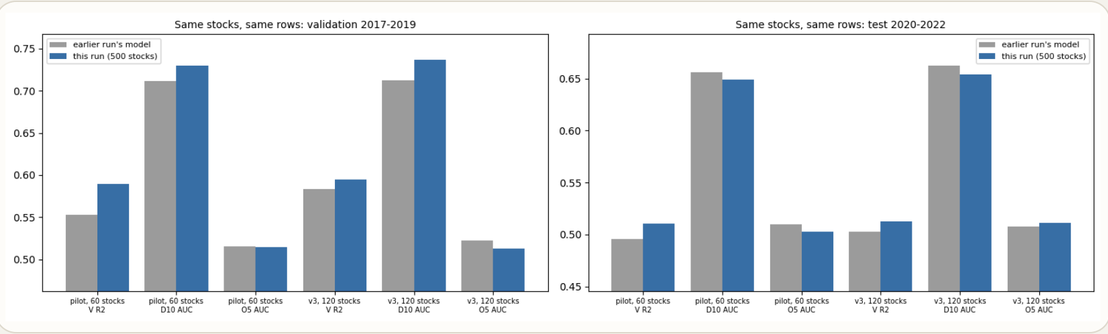
**5 df**
Student-t daily shocks: fat tails chosen on validation

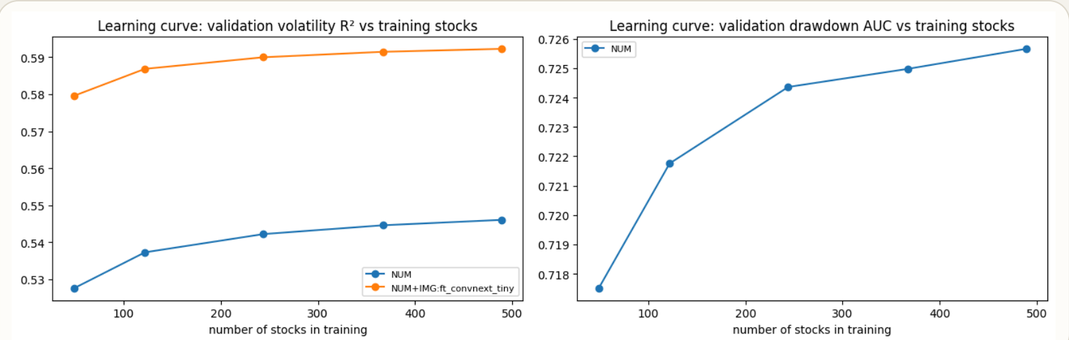
**0 drift**
no direction edge, so no assumed trend

# More stocks helped, with diminishing returns

Earlier runs against this run, on the earlier runs' own stocks and exactly the same test rows:

| Run | Metric | Earlier | 500-stock | Change |
|---|---|---|---|---|
| Pilot (60) | Volatility R² | 0.496 | 0.510 | +0.014 |
| v3 (120) | Volatility R² | 0.503 | 0.513 | +0.010 |
| Pilot (60) | Drawdown AUC | 0.656 | 0.649 | −0.007 |
| v3 (120) | Drawdown AUC | 0.663 | 0.654 | −0.009 |




Validation score against the number of training stocks.

Numbers-only volatility R² rose from 0.528 with 49 stocks to 0.546 with 489, and flattens after about 250. More of the same data will not move R² much further.

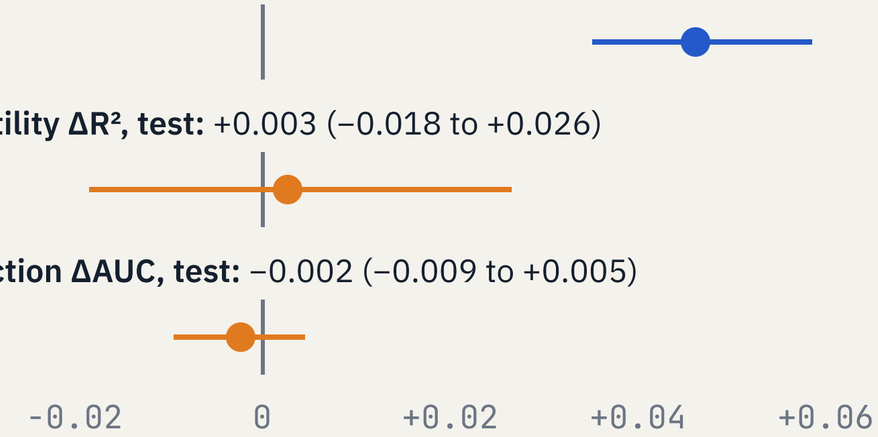
# Images help on validation, not on test

Gain from adding the three chart views to the 33 numbers (95% month-block CIs):

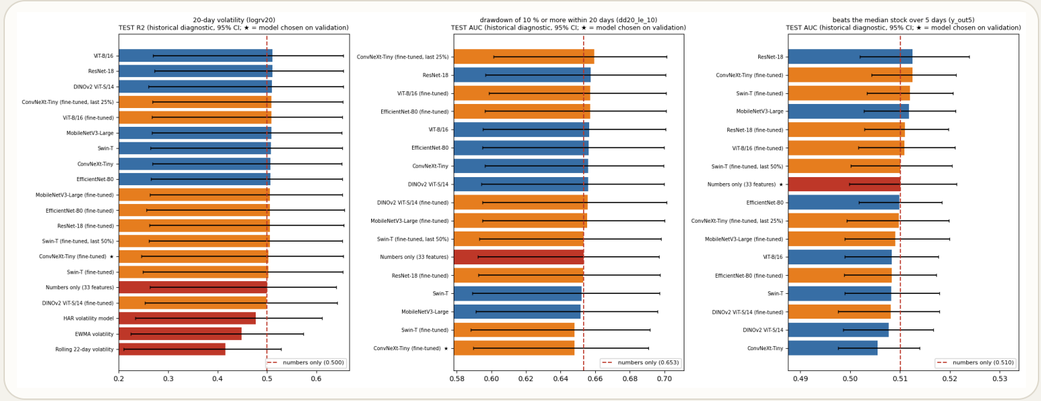
**Volatility ΔR², validation:** +0.046 (+0.035 to +0.059)

**Volatility ΔR², test:** +0.003 (−0.018 to +0.026)

**Direction ΔAUC, test:** −0.002 (−0.009 to +0.005)



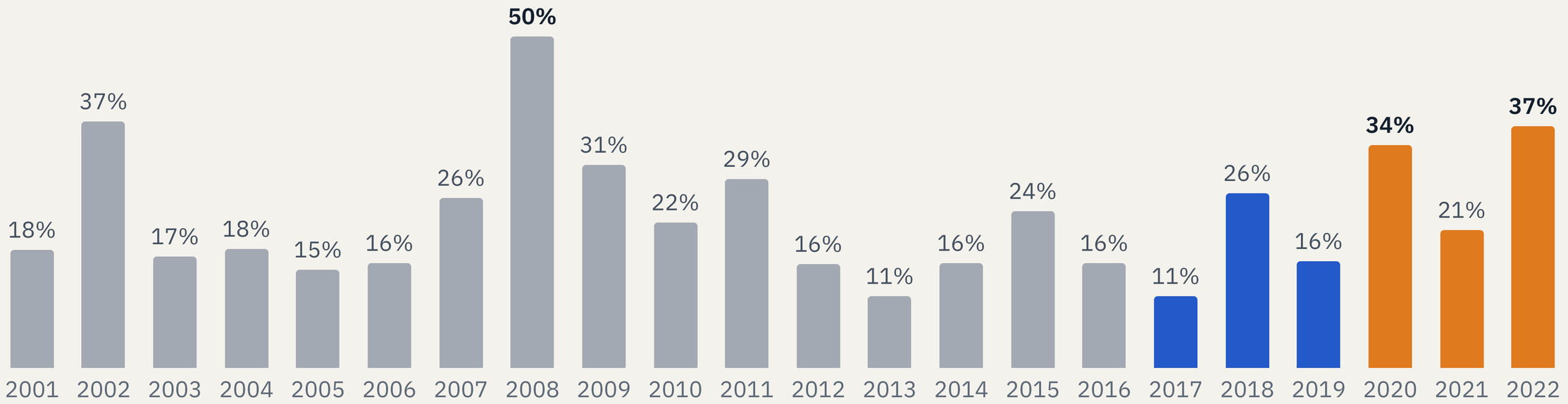

The pictures carry real signal (best image-only R² 0.207 vs about −0.02 for random weights), but it overlaps the 33 numbers, and the renderer hides the volatility level.




Test leaderboard: all candidates overlap within their CIs; the model chosen on validation (★) sits mid-pack.

# The drawdown rate swings from 11% to 50%

Share of stock-days with a 10% drawdown, by year: train validation test




### 2020: the COVID crash

Volatility $R^2$ fell to 0.27, against 0.63 in 2021 and 0.57 in 2022. A market-wide shock no single stock's history could foresee.

### Calm validation years

2017–2019 held no crisis (17.8% drawdowns vs 30.2% in test), so selection could not reward crisis skill.

# Four biases we name rather than hide

### Survivorship

The universe holds only stocks that still trade, so risk is probably understated for firms that later failed.

### Base-rate shift

The drawdown rate rose from 17.8% in validation to 30.2% in test, and calibration error rose to 0.058.

### Selection luck

Picking from many models on one 3-year window rewards noise: the chosen drawdown model ranked last of 17 on test, all within 0.012 AUC.

### Reused test years

Earlier runs read 2020–2022, so the test is a historical check. A clean test needs data after October 2026.

One metric needs care: direction’s F1 of 0.67 at recall 1.00 is what flagging every stock scores, not skill.

# Designed against misuse, open about limits

## Misuse risk

- NO TRADE is the default decision
- Direction calls stay hidden: no edge
- Plain-language explanations and risk tiers
- Labelled research output, not investment advice

## Uneven errors

- Errors peak in crises, when users most need accuracy
- Survivorship may make fragile firms look safer
- Calibration helps; it does not replace judgement

## Transparency and cost

- Code, reports and metric provenance are public
- Data and weights stay private: licence not cleared
- About 4.5 A100-hours for the whole run
- Uploaded screenshots are read in memory, never stored

Claims that AI “reads charts” spread easily, while null results rarely get published. We publish ours.

CONCLUSION

# What we learned, and what comes next

## ESTABLISHED

- Risk is predictable: volatility $R^2$ 0.503, drawdown AUC 0.648
- Beats HAR, EWMA and rolling volatility on the test years
- Fine-tuning helps only when the whole network adapts
- Convex heads certified at their global minimum
- Chart images add no significant test gain; no 5-day edge

## NEXT

- Realised volatility from hourly bars as numeric features
- Walk-forward evaluation that includes 2008 and 2020
- Average the top networks instead of one winner
- A probability floor and rolling recalibration
- A fresh test on data after October 2026

Thank you · github.com/harshatangirala/triframe-cv